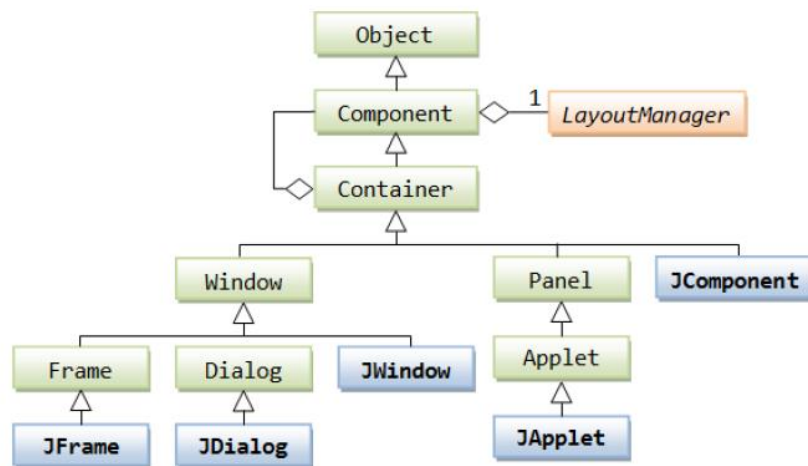


# OBJETOS CONTENEDORES



## abstract class JComponent

```

Color getBackground()
Color getForeground()
int getHeight()
int getWidth()
void paint(Graphics g)
void repaint() //Llama a Paint
void setBackground(Color c)
void setForeground(Color c)
void setFont(Font f)
setBorder(Border d)
setSize(Dimension d)
setPreferredSize(Dimension d)
setMaximumSize(Dimension d)
setMinimumSize(Dimension d)
setLocation(Point p)
setLocationRelativeTo(Component c)
dispose()
setVisible(boolean)
    
```

## abstract class Container

```

add(Component comp)
add(Component comp, int index)
void remove(Component c)
setLayout(LayoutManager layout)
    
```

## class JFrame

Constantes: EXIT\_ON\_CLOSE, DISPOSE\_ON\_CLOSE, DO\_NOTHING\_ON\_CLOSE, HIDE\_ON\_CLOSE

Constructores:

```

JFrame()
JFrame(String title)
    
```

Container getContentPane()

```

void setJMenuBar(JMenuBar menubar) void
setLayout(LayoutManager manager) void
setBounds(int x, int y, int width, int height)
setTitle(String title)
setDefaultCloseOperation(int op)
    
```

## class JDialog

Constructores:

```

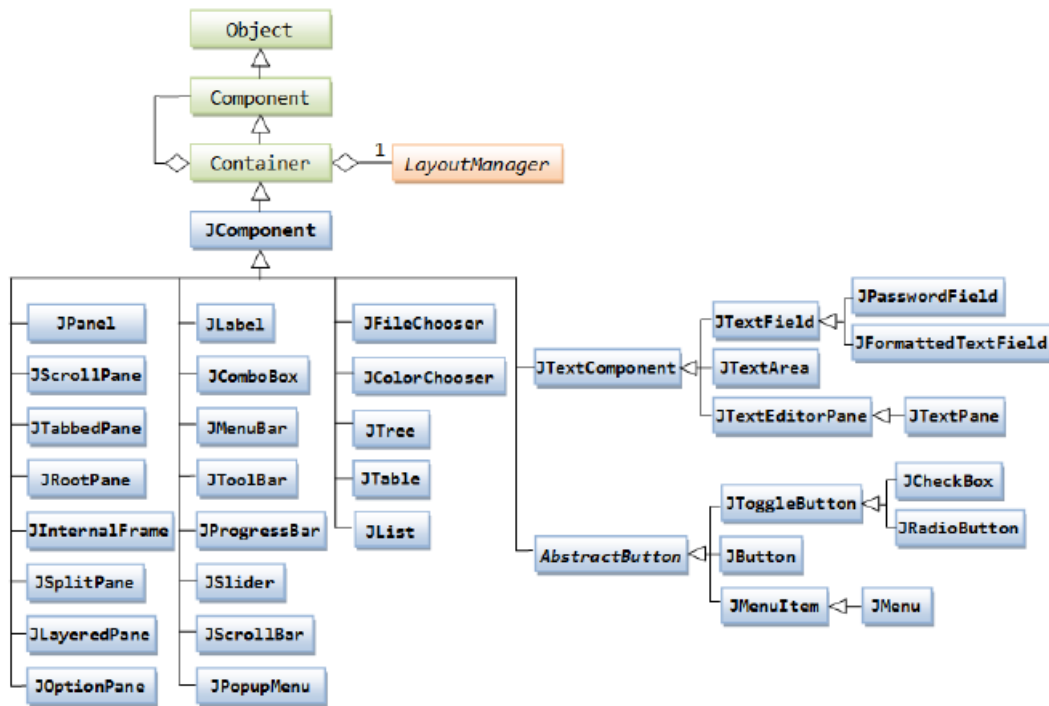
JDialog()
JDialog(Window padre)
JDialog(Window padre, boolean modal)
JDialog(Window padre, String titulo)
JDialog(Window padre, String titulo, boolean modal)
JDialog(Window padre, String titulo, boolean modal, GraphicsConfiguration gc)
    
```

## abstract class Graphics

```

drawLine(int x1, int y1, int x2, int y2) void
drawArc(int x, int y, int width, int height, int startAngle, int arcAngle)
void drawOval(int x, int y, int width, int height)
void drawRect(int x, int y, int width, int height)
void drawString(String str, int x, int y)
void fillArc(int x, int y, int width, int height, int startAngle, int arcAngle)
void fillOval(int x, int y, int width, int height)
Color getColor() Font getFont()
void setColor(Color c)
void setFont(Font f)
    
```

# COMPONENTES



## abstract class JComponent

```

void setPreferredSize(Dimension
preferredSize)
// usar new Dimension(width,
height)
addXXXListener(XXXListener l)

```

## abstract class JPanel

Constructores:

```

JPanel()
JPanel(LayoutManager layout)

```

## class JTextComponent

```

String getText()
void setEditable(boolean b)
void setText(String t)
String getText()

```

## class JTextField

Constantes: LEFT, CENTER, RIGHT

Constructores:

```

JTextField() JTextField(int
columns)
JTextField(String text)
JTextField(String text, int
columns)
void setHorizontalAlignment(int
alignment)

```

## class JTextArea

Constructores:

```

JTextArea()
JTextArea(int rows, int
columns)
JTextArea(String text)
JTextArea(String text, int
rows, int columns)
void append(String str)
void setLineWrap(boolean wrap)

```

## class ImageIcon implements Icon

Constructor:

```

ImageIcon(Image image)
//La imagen se obtiene con getClass().getResource(String path)

```

## class JFileChooser

Constantes: APPROVE\_OPTION, CANCEL\_OPTION, ERROR\_OPTION, FILES\_ONLY, DIRECTORIES\_ONLY, FILES\_AND\_DIRECTORIES

Constructor:

```

JFileChooser()
showOpenDialog(Window padre)
showSaveDialog(Window padre)
getSelectedFile()
setFileSelectionMode(int mode)
setCurrentDirectory(String ruta)
setFileFilter(FileFilter)

```

## class FileNameExtensionFilter implements FileFilter

Constructor:

```

FileNameExtensionFilter(String descripción, String filtro)

```

## class JColorChooser

Método estático:

```

Color showDialog(Component padre, String titulo,Color
inicial)

```

| <b>class JLabel</b>                                                                                                                                                                                                         |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Constantes: LEFT, CENTER, RIGHT                                                                                                                                                                                             |
| Constructores:<br>JLabel()<br>JLabel(String text)<br>JLabel(String text,<br>inhorizontalAlignment)<br>JLabel(Icon icon)<br>JLabel(String text, Icon icon)<br>String getText()<br>void setHorizontalAlignment(int alignment) |

| <b>class JList</b>                                                                                                                                                                                                        |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Constructor:<br>JList()<br>JList(Object[])<br>setListData(Object[] data)<br>setVisibleRowCount(int rows)<br>Object[] getSelectedValues()<br>Object getSelectedItem()<br>int getSelectedIndex()<br>setSelectedIndex(int i) |

| <b>class JComboBox</b>                                                                                                                                                                |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Constructor:<br>JComboBox()<br>JComboBox(Object[])<br>addItem(Object item)<br>Object getSelectedItem()<br>int getSelectedIndex()<br>setSelectedIndex(int i)<br>setEditable(boolean e) |

| <b>abstract class<br/>AbstractButton</b>                                                      |
|-----------------------------------------------------------------------------------------------|
| void addActionListener<br>(ActionListener l)<br>String getText()<br>void setText(String text) |

| <b>class JButton</b>                                                                                         |
|--------------------------------------------------------------------------------------------------------------|
| Constructores:<br>JButton()<br>JButton(String text)<br>JButton(Icon icon)<br>JButton(String text, Icon icon) |

| <b>class JCheckBox</b>                                                                                                                                                                                                  |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Constructor:<br>JCheckBox(String)<br>JCheckBox(String,boolean)<br>JCheckBox(Icon)<br>JCheckBox(Icon,boolean)<br>JCheckBox(String,icon)<br>JCheckBox(String,icon,boolean)<br>setSelected(boolean)<br>isSelected(boolean) |

| <b>class JRadioButton</b>                                                                                                                                                                                                                 |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Constructor:<br>JRadioButton(String)<br>JRadioButton(String,boolean)<br>JRadioButton(Icon)<br>JRadioButton(Icon,boolean)<br>JRadioButton(String,icon)<br>JRadioButton(String,icon,boolean)<br>setSelected(boolean)<br>isSelected(boolean) |

| <b>class ButtonGroup</b>                                 |
|----------------------------------------------------------|
| Constructor:<br>JButtonGroup(String)<br>add(Component c) |

| <b>class JToolBar</b>                                                                                       |
|-------------------------------------------------------------------------------------------------------------|
| Constantes: HORIZONTAL,VERTICAL                                                                             |
| Constructor:<br>JToolBar()<br>JToolBar(int orientation)<br>add(AbstractButton b)<br>setFloatable(boolean b) |

| <b>class JMenu</b>                                                                                           |
|--------------------------------------------------------------------------------------------------------------|
| Constructor:<br>JMenu()<br>JMenu(String text)<br>add(JMenuItem mi)<br>remove(JMenuItem mi)<br>addSeparator() |

| <b>class JMenuBar</b>                      |
|--------------------------------------------|
| Constructor:<br>JMenuBar()<br>add(JMenu c) |

| <b>class JMenuItem</b>                                                                          |
|-------------------------------------------------------------------------------------------------|
| Constructor:<br>JMenuItem()<br>JMenuItem(String text)<br>JMenuItem(String text,<br>ImageIcon i) |

| <b>class JScrollPane</b>                                                                                                                                                                                 |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Constantes:<br>VERTICAL_SCROLLBAR_AS_NEEDED,<br>HORIZONTAL_SCROLLBAR_AS_NEEDED,<br>VERTICAL_SCROLLBAR_NEVER,<br>HORIZONTAL_SCROLLBAR_NEVER,<br>VERTICAL_SCROLLBAR_ALWAYS,<br>HORIZONTAL_SCROLLBAR_ALWAYS |
| Constructor:<br>JScrollPane()<br>JScrollPane(int h,int v)<br>JScrollPane(Component c)<br>JScrollPane(Component c, int h,<br>int i)                                                                       |

| <b>class JSlider</b>                                                                                                                                                                                                                                                                                               |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Constantes: VERTICAL,<br>HORIZONTAL                                                                                                                                                                                                                                                                                |
| Constructor:<br>JSlider()<br>JSlider(int max,int min)<br>JSlider(int max, int min, int valor)<br>JSlider(int orientación, int max, int min, int valor)<br>setMajorTickSpacing(int freq)<br>setMinorTickSpacing(int freq)<br>setPaintTicks(boolean b)<br>setPaintLabels(boolean b)<br>boolean getValueIsAdjusting() |

| <b>class JPopupMenu</b>                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------|
| Constructor:<br>JPopupMenu()<br>add(JMenuItem mi)<br>remove(JMenuItem mi)<br>addSeparator()<br>show(Component donde,int x,int y) |

| <b>class Color</b>                                                                                         |
|------------------------------------------------------------------------------------------------------------|
| Constantes: black, blue, cyan, darkGray, gray, green, lightGray, magenta, orange, pink, red, white, yellow |
| Constructor:<br>Color(int red, int green, int blue)                                                        |

| <b>class Font</b>                                      |
|--------------------------------------------------------|
| Constantes: BOLD, ITALIC, PLAIN                        |
| Constructor:<br>Font(String name, int style, int size) |

# GESTORES DE DISTRIBUCIÓN

|                                                                                                                                                                                          |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>class FlowLayout</b>                                                                                                                                                                  |
| <b>Constantes: CENTER, LEFT, RIGHT</b>                                                                                                                                                   |
| Constructores:<br>FlowLayout()<br>FlowLayout(int align)<br>FlowLayout(int align, int hgap, int vgap)<br>void setAlignment(int align)<br>void setHgap(int hgap)<br>void setVgap(int vgap) |

|                                                                                                                                                            |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>class GridLayout</b>                                                                                                                                    |
| Constructores:<br>GridLayout(int rows, int cols)<br>GridLayout(int rows, int cols, int hgap, int vgap)<br>void setHgap(int hgap)<br>void setVgap(int vgap) |

|                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------|
| <b>class BorderLayout</b>                                                                                               |
| <b>Constantes: CENTER, EAST, NORTH, SOUTH, WEST</b>                                                                     |
| constructores:<br>BorderLayout()<br>BoderLayout(int hgap, int vgap)<br>void setHgap(int hgap)<br>void setVgap(int vgap) |

|                                                                                                                |
|----------------------------------------------------------------------------------------------------------------|
| <b>class BoxLayout</b>                                                                                         |
| <b>Constantes: Y_AXIS,X_ASIS</b>                                                                               |
| Constructores:<br>BoxLayout(Container c, int distribution)<br>void setHgap(int hgap)<br>void setVgap(int vgap) |

## EVENTOS

|                                      |
|--------------------------------------|
| <b>abstract class ActionListener</b> |
| void actionPerformed(ActionEvent e)  |

|                                                 |
|-------------------------------------------------|
| <b>interface Event</b>                          |
| Object getSource()<br>String getActionCommand() |

|                                                                                                                                                                                                                                                                           |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>abstract class WindowListener</b>                                                                                                                                                                                                                                      |
| void windowActivated(WindowEvent e)<br>void windowClosed(WindowEvent e)<br>void windowClosing(WindowEvent e)<br>void windowDeactivated(WindowEvent e) void<br>windowDeiconified(WindowEvent e)<br>void windowIconified(WindowEvent e)<br>void windowOpened(WindowEvent e) |

|                                                                                                                                                                             |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>abstract class MouseListener</b>                                                                                                                                         |
| void mouseClicked(MouseEvent e)<br>void mouseEntered(MouseEvent e)<br>void mouseExited(MouseEvent e)<br>void mousePressed(MouseEvent e)<br>void mouseReleased(MouseEvent e) |

|                                                                                                                                                                         |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>abstract class MouseEvent</b>                                                                                                                                        |
| Consantes: NOBUTTON, BUTTON1, BUTTON2, BUTTON3                                                                                                                          |
| int getX()<br>int getY()<br>int getXOnScreen()<br>int getYOnScreen()<br>int getLocationOnScreen()<br>int getButton()<br>int getClickCount()<br>boolean isPopUpTrigger() |

|                                                                |
|----------------------------------------------------------------|
| <b>abstract class FocusListener</b>                            |
| void focusGained(FocusEvent e)<br>void focusLost(FocusEvent e) |

|                                                                                          |
|------------------------------------------------------------------------------------------|
| <b>abstract class KeyListener</b>                                                        |
| void keyPressed(KeyEvent e)<br>void keyReleased(KeyEvent e)<br>void keyTyped(KeyEvent e) |

|                                                         |
|---------------------------------------------------------|
| <b>abstract class KeyEvent</b>                          |
| <b>Contantes: Códigos correspondientes a las teclas</b> |
| char getKeyChar()<br>int getKeyCode()                   |

|                                      |
|--------------------------------------|
| <b>abstract class ChangeListener</b> |
| void stateChanged(ChangeEvent e)     |

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>class JOptionPane</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| Constantes: ERROR_MESSAGE, INFORMATION_MESSAGE, WARNING_MESSAGE, QUESTION_MESSAGE, PLAIN_MESSAGE, YES_NO_OPTION, YES_NO_CANCEL_OPTION, OK_CANCEL_OPTION                                                                                                                                                                                                                                                                                                                                                                                                                             |
| Métodos estáticos:<br>void showMessageDialog(Component padre, Object msg, Object titulo [op], int tipo [op], Icon icono [op])<br>int showConfirmDialog(Component padre, Object msg, Object titulo [op], int botones [op], int tipo [op], Icon icono [op]<br>String showConfirmDialog(Component padre , Object msg, Object titulo [op], int tipo [op], Icon icono [op], Object[] valores, Object valorinicial)<br>String showOptionDialog(Component padre , Object msg, Object titulo [op], int botones, int tipo, Icon icono [op], Object[] titulosBotones, Object botonPorDefecto) |